

Computer Science & IT

Discrete Mathematics



Mathematical Logic

Lecture No. 02



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Recap of Previous Lecture



✓
Topic

Propositions and their types

✓
Topic

Connectives

✓
Topic

Tautology, Contradiction, Contingency and Satisfiable propositional functions

Topics to be Covered



✓
Topic

Logical Implication

✓
Topic

Logical Equivalences

✓
Topic

Argument / Inference

✓
Topic

Rules of inference

✓
Topic

Practice questions

Mathematical Logic

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graph TD; A[Mathematical Logic] --> B[① Propositional Logic]; A --> C[② Predicate Logic];
```

✓
① Propositional Logic

✗
② Predicate Logic



Topic : Propositional Logic

* Proposition : —

A declarative sentence to which we can assign only one of the truth values (i.e. TRUE/FALSE) is called a propositional statement or proposition.

- eg:
- It rains today
 - Cricket match is played.



Topic : Assumptions of proposition

Law of excluded middle : If a proposition is not true then it will be false, and similarly if a proposition is not false then it will be true.

Law of contradiction : A proposition can not be true as well as false simultaneously.



Topic : Atomic propositions

Atomic propositions are simple propositions which can not be divided further.

eg: ① It rains today

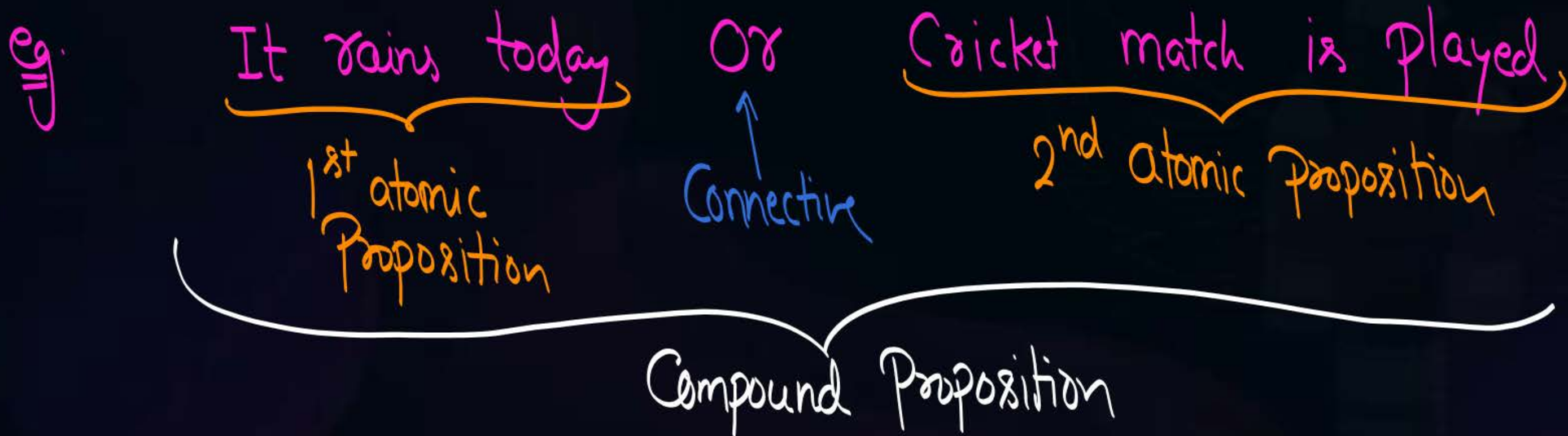
② Cricket match is played.



Topic : Compound Propositions

- Compound propositions are constructed by combining atomic propositions with the help of connectives.

Compound propositions are also known as propositional formula, or propositional function.





Topic : Connectives

- ✓ 1. Negation ($\neg/\sim$) / Not
- ✓ 2. AND ($\wedge$) / Conjunction
- ✓ 3. OR ($\vee$) / Disjunction
- ✓ 4. Implication ($\rightarrow$)
- ✓ 5. Biconditional ($\leftrightarrow$)

Imp.



Topic : Negation ($\neg/\sim$)

- If p is any proposition, then negation of p / $\text{not}(p)$ / $\neg p$ is also a proposition whose truth value is true only when p is false and it is false only when p is true.

p	$\sim p$
T	F
F	T



Topic : AND ($\wedge$) / Conjunction

If p and q are any two propositions, then “ p AND q ” / $p \wedge q$ is also a proposition whose truth value is true only when both p as well as q are true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F



Topic : OR ($\vee$) / Disjunction

If p and q are any two propositions, then " p OR q " / $p \vee q$ is also a proposition whose truth value is false only when both p as well as q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

If it rains today then Cricket match will not be played

p q

if p then q

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1



Topic : Implication ($\rightarrow$)

aka (if-then)

If p and q are any two propositions, then p implies q (i.e. $p \rightarrow q$ / if p then q) is also a proposition whose truth value is false only when p is true, and q is false.

$p \rightarrow q$
Antecedent Consequent

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

" $P \rightarrow Q$ " & " $\sim P \vee Q$,"
have same truth table
∴ $P \rightarrow Q \equiv \sim P \vee Q$

We can observe
 $Q \rightarrow P \equiv \sim P \rightarrow \sim Q$

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim P \vee Q$	$Q \rightarrow P$	$\sim P \rightarrow \sim Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T	T	T	T
T	F	F	T	F	F	F	F	F
F	T	T	F	T	T	T	T	F
F	F	T	T	T	T	F	F	T

$P \rightarrow Q \equiv \sim Q \rightarrow \sim P$



Topic : Implication ($\rightarrow$)

- ❑ If p is true, then q must be true for $p \rightarrow q$ to be true.
- ❑ If p is false, then $p \rightarrow q$ is *always* true whatever is the truth value of q .
- ❑ If q is true, the $p \rightarrow q$ is *always* true whatever is the truth value of p .
- ❑ Truth table of $p \rightarrow q$ and truth table of $\neg p \vee q$ are exactly same, therefore both are equivalent.
- ❑ Converse of $p \rightarrow q$ is $q \rightarrow p$.
- ❑ Opposite or Inverse of $p \rightarrow q$ is " $\neg p \rightarrow \neg q$ "
- ❑ Contra positive for $p \rightarrow q$ is " $\neg q \rightarrow \neg p$ "

$$p \rightarrow q \equiv \sim p \vee q$$

$$p \rightarrow q \equiv \underbrace{\sim q \rightarrow \sim p}_{\text{Contra positive of } p \rightarrow q}$$

$$p \rightarrow q \not\equiv \sim p \rightarrow \sim q$$

(inverse of $p \rightarrow q$)

$$p \rightarrow q \not\equiv q \rightarrow p$$

(converse of $p \rightarrow q$)

$$q \rightarrow p \equiv \sim p \rightarrow \sim q$$

Converse of $p \rightarrow q$ is
equivalent to inverse of $p \rightarrow q$



Topic : Biconditional ($\leftrightarrow$)

(if and only if)

- If p and q are any two propositions, then " $p \leftrightarrow q$ " (i.e. p if and only if q) is also a proposition whose truth value is true only when both p and q have same truth value.

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

We can observe
 $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$



Topic : Biconditional ($\leftrightarrow$)

We observed $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

" $p \leftrightarrow q$ " is true if and only if
both " $p \rightarrow q$ " and " $q \rightarrow p$ " are true



Topic : Tautology

Tautology is term which is defined w.r.t. propositional logic



A propositional function which is always true is called tautology.

A propositional function which is always true is also called a valid propositional function.

W.r.t. propositional logic,

Tautologies & Valid propositional function
are same things

But it will
not be the case
w.r.t. predicate
logic

eg.

$$p \vee \sim p \quad f(p)$$

p	$\sim p$	$p \vee \sim p$
---	----------	-----------------

T

F

T

F

T

T

} Always true
∴ Tautology

eg.

Check if $p \rightarrow (q \rightarrow p)$ is a tautology or not

p	q	$(q \rightarrow p)$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

} Always True

∴ $p \rightarrow (q \rightarrow p)$ is a tautology



Topic : Contradiction



- A propositional function which is always false is called Contradiction.

$f(p)$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

} Always false
∴ Contradiction

Note:-

① If a propositional function is always true then it is called Valid propositional function

② If propositional function is false for at least one case, then it is an invalid propositional function

✶ Each Contradiction is invalid propositional function, but Each invalid propositional function need not be Contradiction



Topic : Contingency



A propositional function which is neither a tautology nor a contradiction is called a contingency.

i.e., A contingency is true for at least one case as well as false for at least one case.

eg:-

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$f(p, q)$

True for at least one Combination
as well as

false for at least one Combination

so it is a Contingency.



Topic : Satisfiable



A propositional function which is true for at least one case is called a satisfiable propositional function.

- i.e., if a propositional function is not satisfiable, then it is a Contradiction
- if propositional function is not a Contradiction, then it is satisfiable



Topic : NOTE



- Every propositional function which is not a contradiction is satisfiable.
- Every tautology is satisfiable but not vice-versa.
i.e. Every Satisfiable propositional function, need not be a tautology
- Every contingency is satisfiable, but every satisfiable function need not be contingency.
→ eg: Tautology — it is Satisfiable, but not a Contingency.



Topic : NOTE



Tautology	Contradiction	Contingency
✓ Always True	✓ Always False	✓ Neither always true nor always false
✓ Valid	✓ Invalid	✓ Invalid
✓ Satisfiable	✓ Not satisfiable	✓ Satisfiable

Today's Topic



Topic : Logical Implication / Implication

- Let P and Q are any two propositional functions.
- Whenever P is true if Q is also true then P logically implies Q .
and if it is given that P logically implies Q , then whenever P is true Q is guaranteed to be true
- P logically implies Q if and only if $P \rightarrow Q$ is a tautology.

S1: "P logically implies Q"

Meaning: $P \rightarrow Q$ is a tautology or Whenever P is true
Q is guaranteed to be true

Q:- Check whether "P logically implies Q" is valid or invalid

• If $P \rightarrow Q$ is a tautology, then "P logically implies Q" is valid

• If $P \rightarrow Q$ is not a tautology, then "P logically implies Q" is invalid



Topic : Logical Equivalence / Equivalence

- If P and Q are any two propositional functions, then P equivalent to Q is written as $P \equiv Q$. or $P \iff Q$
- P and Q are said to be equivalent if and only if they have same truth table.
- $P \equiv Q$ if and only if $P \iff Q$ is a tautology.
- $P \equiv Q$ if and only if, $P \rightarrow Q$ is a tautology as well as $Q \rightarrow P$ is a tautology.
- $P \equiv Q$ if and only if, P logically implies Q as well as Q logically implies P



Topic : Some important equivalences

① $\rightarrow \sim(\sim P) \equiv P$ } Double Complementation

② $\rightarrow \left. \begin{array}{l} P \vee Q \equiv Q \vee P \\ P \wedge Q \equiv Q \wedge P \end{array} \right\}$ Commutative

③ $\rightarrow \left. \begin{array}{l} (P \vee Q) \vee R \equiv P \vee (Q \vee R) \\ (P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R) \end{array} \right\}$ Associative



Topic : Some important equivalences

$$\textcircled{4} \quad P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

Distributive Property

$$\textcircled{5} \quad \sim(P \wedge Q) \equiv (\sim P \vee \sim Q)$$

$$\sim(P \vee Q) \equiv (\sim P \wedge \sim Q)$$

De' Morgan law

$$\textcircled{6} \quad P \wedge (P \vee Q) \equiv P$$

$$P \vee (P \wedge Q) \equiv P$$

Absorption law



Topic : Some important equivalences

$$\begin{array}{l} \textcircled{7} \quad P \wedge P \equiv P \\ \quad \quad P \vee P \equiv P \end{array} \quad \left. \vphantom{\begin{array}{l} P \wedge P \equiv P \\ P \vee P \equiv P \end{array}} \right\} \text{Idempotent law}$$

$$\begin{array}{l} P \vee T \equiv T \\ P \wedge F \equiv F \end{array} \quad \left. \vphantom{\begin{array}{l} P \vee T \equiv T \\ P \wedge F \equiv F \end{array}} \right\} \text{Domination law}$$

$$\begin{array}{l} P \vee F \equiv P \\ P \wedge T \equiv P \end{array} \quad \left. \vphantom{\begin{array}{l} P \vee F \equiv P \\ P \wedge T \equiv P \end{array}} \right\} \text{Identity}$$

$$T \wedge T \equiv T$$

$$T \vee T \equiv T$$

$$T \wedge F \equiv F$$

$$T \vee F \equiv T$$

$$F \vee F \equiv F$$

$$F \wedge F \equiv F$$

+ Some important Equivalences :-

①

$$P \rightarrow Q \equiv \sim P \vee Q$$

②

$$P \rightarrow Q \equiv \sim Q \rightarrow \sim P$$

③

$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

H.W

Q. Simplify the following Statement

$$\{\sim P \wedge (\sim q \wedge r)\} \vee \{(q \wedge r) \vee (P \wedge r)\}$$

- (a) True
- (b) False
- (c) p
- (d) q
- (e) r

H.W.
Q: Simplify the following Expression

$$\left[(P \vee Q) \wedge \sim \{ \sim P \wedge (\sim Q \vee \sim R) \} \right] \vee \left[(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R) \right]$$

H.W. Q: Which of the following is/are valid?

- a) $(P \wedge Q)$ logically implies $(P \leftrightarrow Q)$
- b) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow Q)$
- c) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow \sim Q)$
- d) Q logically implies $(P \rightarrow Q)$

H.W. Q: Which of the following is/are tautology?

a) $(P \wedge Q) \longrightarrow (P \longleftrightarrow Q)$

b) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow Q)$

c) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow \sim Q)$

d) $Q \longrightarrow (P \longrightarrow Q)$

H.W. Q: Which of the following is/are tautology?

(a) $(a \vee b) \rightarrow (b \wedge c)$

(b) $(a \wedge b) \rightarrow (b \vee c)$

(c) $(a \vee b) \rightarrow (b \rightarrow c)$

(d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$

LHS True then RHS True

or

RHS is false then LHS False

H.W. Q. Consider the following statements

$S_1: \{(A \wedge B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$

$S_2: \{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

then which of the following is true ?

- (a) Both S_1 & S_2 are valid
- (b) Only S_1 is valid
- (c) Only S_2 is valid
- (d) Neither S_1 nor S_2 is valid.

H.W. Q. Which of the following is are true?

(a) $\{a \rightarrow (b \vee c)\} \equiv \{(a \wedge \sim b) \rightarrow c\}$

(b) $\{(P \rightarrow Q) \wedge (R \rightarrow Q)\} \equiv \{(P \vee R) \rightarrow Q\}$

(c) $\{(P \rightarrow Q) \vee (R \rightarrow Q)\} \equiv \{(P \wedge R) \rightarrow Q\}$

(d) All of the above

H.W.

Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above propositional statement is.

- (a) Satisfiable but not valid
- (b) Valid
- (c) Contradiction

Statement S_1 : If it rains today then Cricket match is not played

Statement S_2 : Cricket match is not played

Conclusion: It rained today

→ It is an argument

↓
Given argument
may be Valid or
may be invalid

Our job is to
→ Check whether the
given argument is
Valid or invalid.



Topic : Argument / Inference

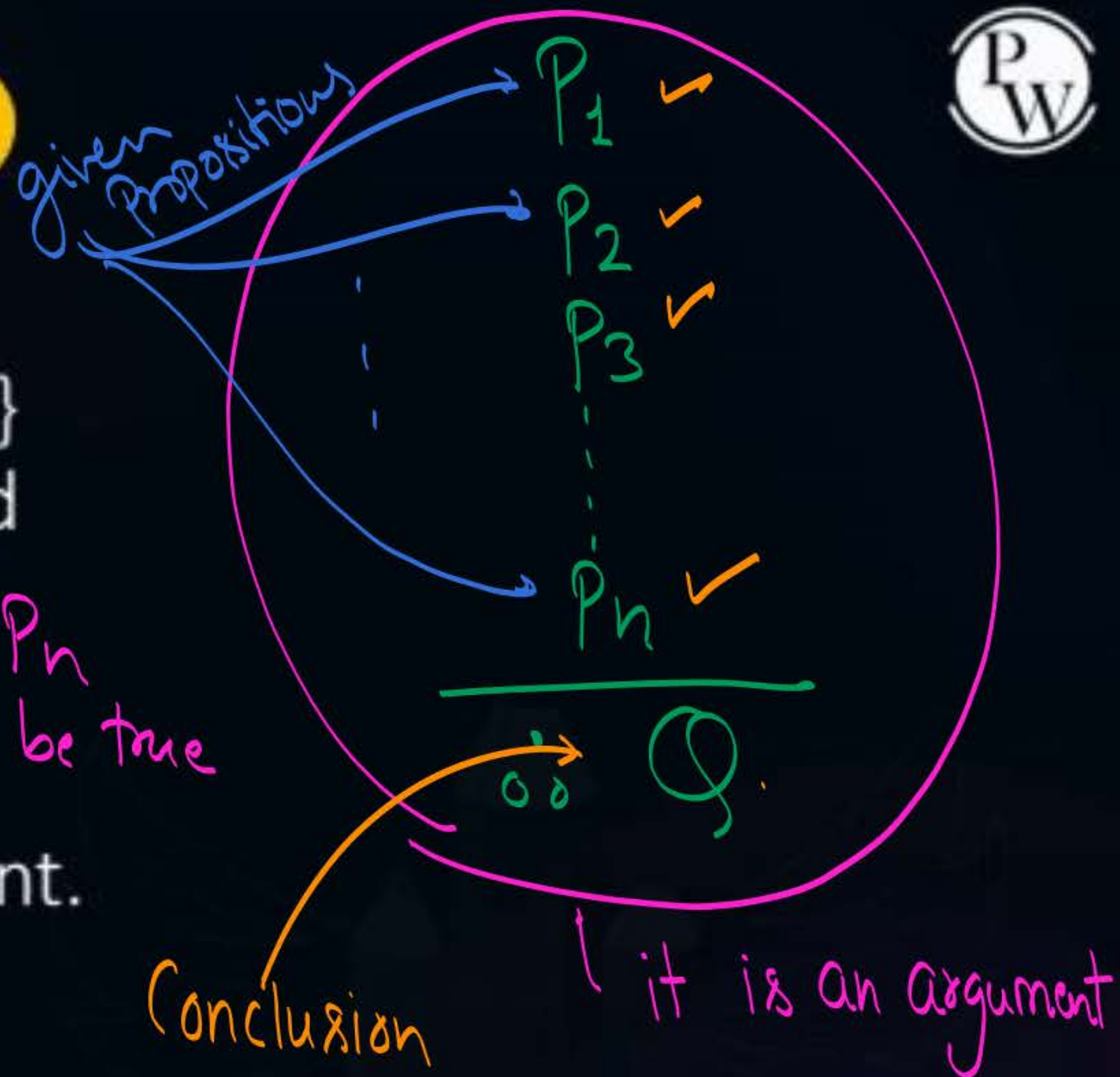


The statement that,

- A set of premises $\{P_1, P_2, P_3, \dots, P_n\}$ yields another proposition Q is called an argument.

The argument is, that whenever $P_1, P_2, \dots, P_n$ are true then Conclusion Q is guaranteed to be true

Q is called conclusion of the argument.





Topic : Argument / Inference



- ✓✱ Argument may be valid or invalid.
- ✓✱ The process of reasoning whether the argument is valid or invalid is called inference.
- ✓✱ If conclusion Q can be inferred from the set of premises by applying some rules of inference and equivalences, then argument is said to be valid otherwise invalid.



Topic : Argument / Inference

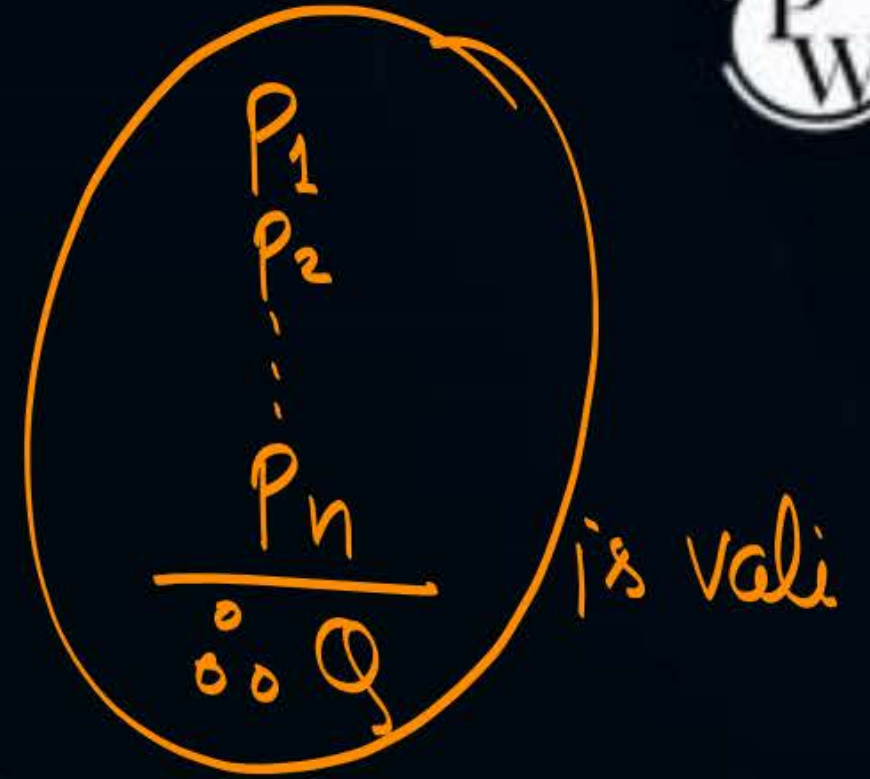
Following statements are equivalent, *yields*

• Argument $\{P_1, P_2, P_3, \dots, P_n\} \vdash Q$ is valid. $\equiv$

• $\{P_1, P_2, P_3, \dots, P_n\}$ Logically implies Q is valid

• $\{P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n\} \rightarrow Q$ is a tautology

• Conclusion Q follows from the premises $\{P_1, P_2, \dots, P_n\}$





Topic : Rules of inference



Any valid *reasoning* is rule of inference.



Topic : Rules of inference



1. Simplification

$$\left\{ \frac{P \wedge Q}{\therefore P} \text{ is valid} \right.$$

Similarly

$$\frac{P \wedge Q}{\therefore Q} \text{ is valid}$$

- * P follows from $P \wedge Q$
- * $(P \wedge Q)$ logically implies P is true
- * $(P \wedge Q) \rightarrow P$ is a tautology
- * $(P \wedge Q) \vdash P$ is valid



Topic : Rules of inference



2. Addition

$\frac{P}{\therefore P \vee Q}$ is valid

- * $P \rightarrow (P \vee Q)$ is a tautology
- * $(P \vee Q)$ follows from P
- * P logically implies $(P \vee Q)$

→ Addition can be performed using disjunction (not using conjunction)

i.e. $\frac{P}{\therefore P \wedge Q}$ is invalid

→ Simplification can be performed w.r.t. Conjunction Only

i.e. $\frac{P \vee Q}{\therefore P}$ is invalid



Topic : Rules of inference



3. Conjunction

$$\begin{array}{l} \text{(i)} \quad P \\ \text{(ii)} \quad Q \\ \hline \therefore P \wedge Q \end{array} \text{ is valid}$$



Topic : Rules of inference



4. Disjunctive Syllogism

$$\frac{P \vee Q \quad \sim P}{\therefore Q} \text{ is valid}$$

$$\frac{P \vee Q \quad \sim Q}{\therefore P} \text{ is Valid}$$



Topic : Rules of inference



5. Conjunctive Syllogism

$$\left. \begin{array}{c} \sim(P \wedge Q) \\ P \end{array} \right\} \text{ is valid}$$

$$\therefore \sim Q$$

$$\begin{array}{l} \text{let } P \wedge Q \text{ is false} \\ \& P \text{ is true} \\ \hline \therefore Q \text{ is false} \end{array}$$



Topic : Rules of inference



6. Modus Ponens

$$\begin{array}{c} P \rightarrow Q \\ P \\ \hline Q \end{array} \text{ is valid}$$

7. Fallacy of affirming the consequent

Mistake

$$\begin{array}{c} P \rightarrow Q \\ Q \\ \hline P \end{array} \text{ is invalid}$$



Topic : Rules of inference



8. Modus Tollen's

$$\begin{array}{l} P \rightarrow Q \\ \sim Q \\ \hline \therefore \sim P \end{array} \text{ is valid}$$

9. Fallacy of denying the antecedent

~~$$\begin{array}{l} P \rightarrow Q \\ \sim P \\ \hline \therefore \sim Q \end{array} \text{ is invalid}$$~~



Topic : Rules of inference



10. Transitivity

$$\begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \\ \hline \therefore P \rightarrow R \end{array} \text{ is valid}$$

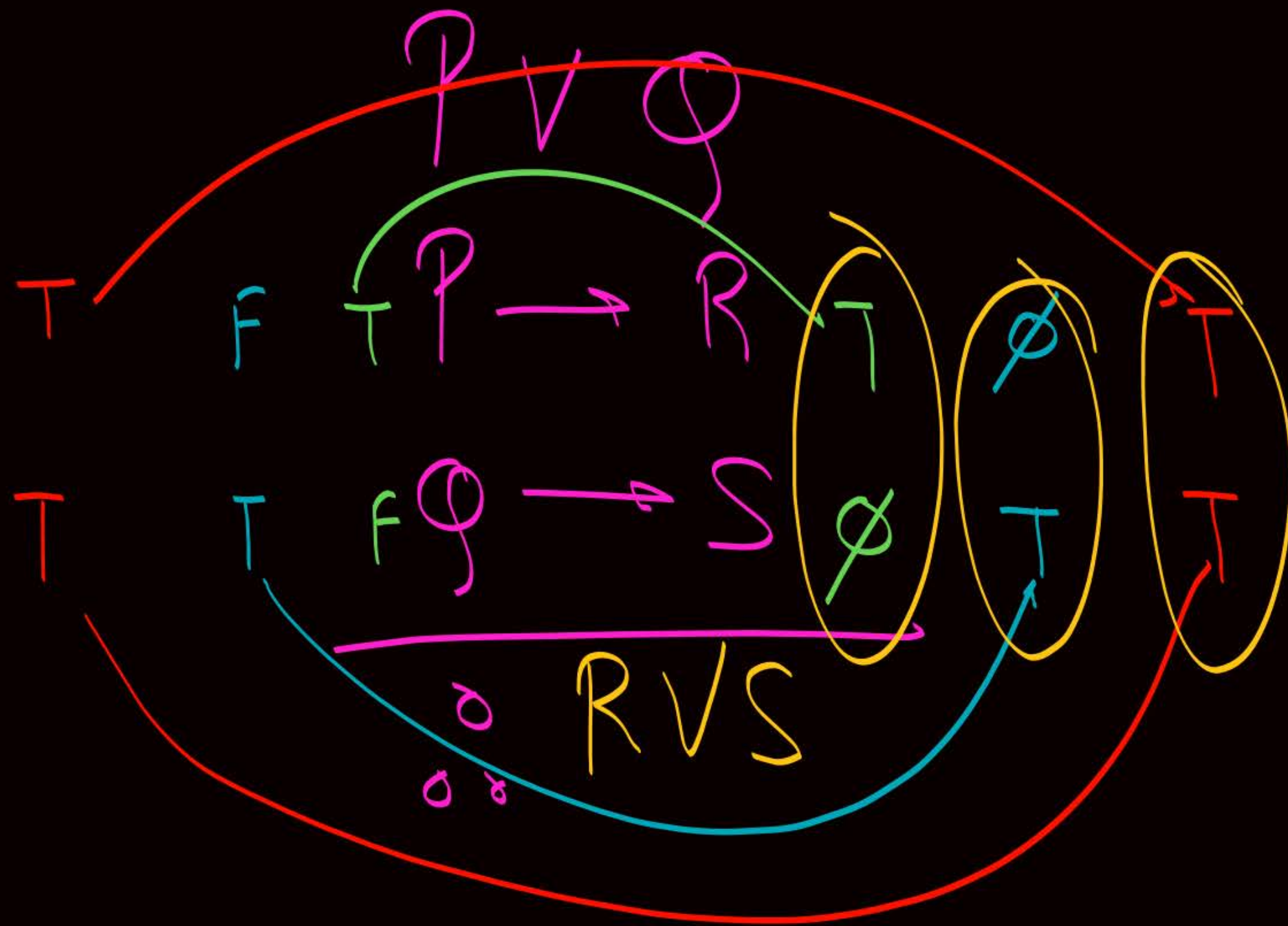


Topic : Rules of inference



11. Dilemma

$$\begin{array}{l} P \vee Q \quad \{ \text{At least one of } P \text{ or } Q \text{ is true} \} \\ P \rightarrow R \\ Q \rightarrow R \\ \hline \therefore R \end{array} \text{ is valid}$$





Topic : Rules of inference



12. Constructive Dilemma

$$\begin{array}{l} P \vee Q \\ P \rightarrow R \\ Q \rightarrow S \\ \hline \therefore R \vee S \end{array}$$

is valid

{ At least one of P or Q is true }

{ then at least one of R or S is guaranteed to be true }

F	X	F	$P \rightarrow R$	F	T	F
X	F	F	$Q \rightarrow S$	T	F	F

$\sim R \vee \sim S \equiv T$ } At least one of R or S is false }

At least one of P or Q will be false $\therefore \text{i.e.} = \sim P \vee \sim Q$



Topic : Rules of inference



13. Destructive Dilemma

$$P \rightarrow R$$

$$Q \rightarrow S$$

$$\sim R \vee \sim S$$

{ At least one of R or S is false }

$$\therefore \sim P \vee \sim Q$$

{ At least one of P or Q is false }



Topic : Rules of inference



14. Resolution

$$p \vee q$$

$$\sim p \vee r$$

$$\hline \therefore q \vee r$$

is valid

$$p \vee q$$
$$\sim p \vee r$$

Exactly '1' of
 p or $\sim p$ is
guaranteed to be
false

↓
Hence at least one
of q or r
must be true



Topic : Rules of inference



15. Non Sequitur

$$\begin{array}{c} p \\ q \\ \hline \therefore r \end{array} \text{ is invalid}$$

Q:- Which of the following is/are true?

(a) Conclusion R follows from the premises.

$$\{ P \rightarrow (Q \rightarrow R), (P \wedge Q) \}$$

(b) $\{ P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P \}$ logically implies S

(c) $\{ (P \vee Q), \sim P \} \vdash (P \wedge Q)$ is valid

(d) $\{ (\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S) \} \rightarrow W$ is a tautology

Q:- Which of the following is/are true?

@ Conclusion R follows from the premises.

$$\{ P \rightarrow (Q \rightarrow R), (P \wedge Q) \}$$

III Simplification

P, Q

M.P.

$$\therefore Q \rightarrow R$$

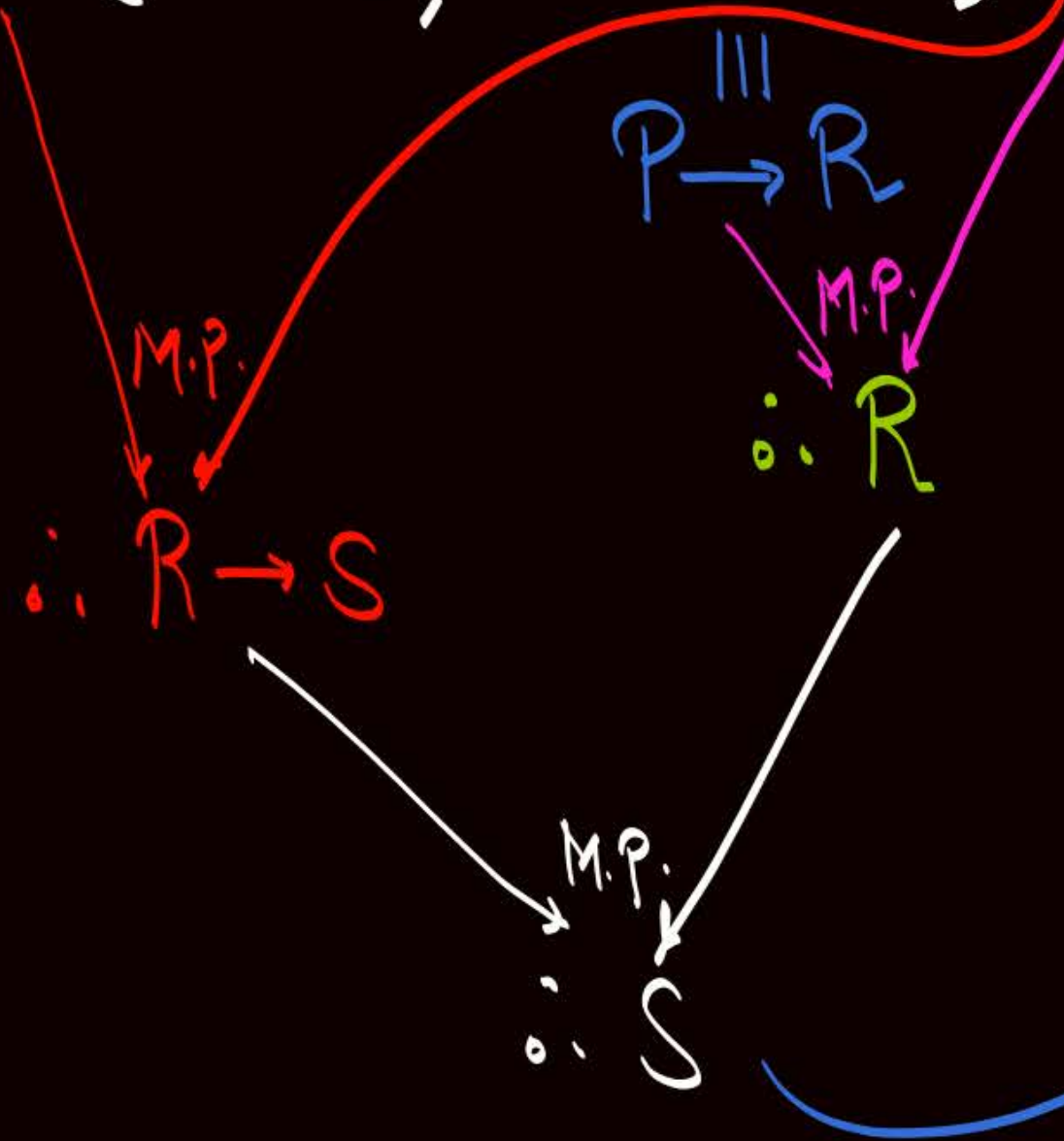
M.P.

$\therefore R$

Hence, R follows from set of Premises

Q:- Which of the following is/are true?

(b) $\{ P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P \}$ logically implies S



Q:- which of the following is/are true?

© $\{ (P \vee Q), \sim P \} \vdash (P \wedge Q)$ is valid

$\swarrow$ D.S. $\searrow$
 $\therefore Q$

$\therefore Q \wedge P$

is invalid
↓
Addition is
not possible
Using Conjunction.

if $\sim P$ is true

i.e. if 'P' is false
then " $P \wedge Q$ " can
never be true

Q:- Which of the following is/are true?

(d) $\{ (\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S) \} \rightarrow W$ is a tautology

$\sim T \rightarrow \sim R, \sim S, T \rightarrow W, R \vee S$

$R \xrightarrow{w} T$

$\therefore R$

$\therefore T$

$\therefore W$



Q:- Which of the following is/are true?

☒ (a) Conclusion R follows from the premises.

$$\{ P \rightarrow (Q \rightarrow R), (P \wedge Q) \}$$

☒ (b) $\{ P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P \}$ logically implies S

☐ (c) $\{ (P \vee Q), \sim P \} \vdash (P \wedge Q)$ is valid

☒ (d) $\{ (\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S) \} \rightarrow W$ is a tautology

H.W. Q:- Check whether the following argument is valid or invalid

- * If today is Vishal's B'day, then today is 1st April
- * Today is 1st April

∴ Today is Vishal's B'day

let P = Today is Vishal's B'day

Q = Today is 1st April

H.W. Q:- Check whether the following argument is valid or invalid

- * If today is Vishal's B'day, then today is 1st April
- * Today is not 1st April

∴ Today is not Vishal's B'day

P = Today is Vishal's B'day

$\sim P$ = Today is not Vishal's B'day

Q = Today is 1st April

$\sim Q$ = Today is not 1st April

H.W.
★

Q: Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies $\sim P$

H.W.

Q:- Check whether the following argument is
Valid or not?

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

Think
about
Solution



2 mins Summary



✓
Topic

Logical Implication

✓
Topic

Logical Equivalences

✓
Topic

Argument / Inference

✓
Topic

Rules of inference

✓
Topic

Practice questions

THANK - YOU